

PhD candidate in mathematical statistics focusing on the theory of causal inference/discovery, extreme value theory and uncertainty quantification.

EDUCATION

- **University of Lausanne (DO-HEC), Switzerland** Sep 2021 – July 2026
PhD candidate under Prof. Valérie Chavez-Demoulin Lausanne, Switzerland
 - Topic: Estimating Unidentifiable Objects in Causal Inference
- **Charles University, Faculty of Mathematics and Physics** 2016 – 2021
BSc & MSc in Pure Mathematics and Mathematical Statistics Prague, Czech Republic
 - Excellence scholarship (grades in top few %)
 - Best Master's Thesis in Statistics (10,000 CZK award) under Prof. Zbyněk Pawlas
 - Research assistant on the Primus GeMS project for outstanding research

EXPERIENCE

- **University of Osaka, Faculty of Data Science** Dec 2025 – Feb 2026
Visiting Student Researcher Osaka, Japan
 - Collaborated with Prof. Shohei Shimizu on causal discovery.
- **UC Berkeley, Department of Statistics** Aug 2024 – Jan 2025
Visiting Student Researcher Berkeley, USA
 - Collaborated with Prof. Bin Yu on causal inference and uncertainty quantification.
- **Czech Academy of Sciences** Jun 2020 – Jun 2021
Research Assistant (part-time) Prague, Czech Republic
 - Conducted research under Prof. Milan Paluš; forming the basis of my master's thesis.

PUBLICATIONS

1. **J. Bodik** (2026). *Counterfactual prediction with cross-world dependence*. **CleaR**.
2. **J. Bodik**, V. Chavez-Demoulin (2025). *Identifiability of causal graphs under nonadditive conditionally parametric causal models*. **Journal of Machine Learning Research**.
3. **J. Bodik**, V. Chavez-Demoulin (2025). *Structural restriction in local causal discovery: identifying direct causes of a target variable*. **Biometrika**.
4. I. Azizi*, **J. Bodik***, J. Heiss*, B. Yu (2025). *CLEAR: calibrated learning for epistemic and aleatoric risks*. **ICLR**. *Co-first authors.
5. **J. Bodik** (2024). *Extreme treatment effect: extrapolating causal effects into extreme treatment domain*. **Mathematics**.
6. **J. Bodik**, Z. Pawlas, M. Paluš (2024). *Causality in extremes of time series*. **Extremes**.

PREPRINTS

1. **J. Bodik**, Y. Huang, B. Yu (2025). *Cross-world assumption and prediction intervals for individual treatment effects*. Preprint submitted to **JRSS B**. [arXiv:2507.12581](#)
2. **J. Bodik**, O. Pasche (2024). *Granger causality in extremes*. Preprint submitted to **Bernoulli**. [arXiv:2407.09632](#)
3. **J. Bodik**, L. Mhalla, V. Chavez-Demoulin (2023). *Detecting causal covariates for extreme dependence structures*. Unpublished semester project; [arXiv:2212.09831](#)

GRANTS

- **Travel Grant FUnil (≈5,000 USD)** Dec 2025 – Feb 2026
Travel grant supporting research visit to Osaka University
 - Visiting Student Researcher in Prof. Shohei Shimizu's group.
- **Hasler Foundation Research Grant (≈10,000 USD)** Aug 2024 – Jan 2025
Highly competitive grant supporting research visit to UC Berkeley
 - Visiting Student Researcher in Prof. Bin Yu's group.

HONORS & AWARDS

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| • Best Master's Thesis Award (10,000 CZK)
<i>Top three master's theses in probability, statistics, and econometrics at Charles University</i> | 2021 |
| • Research Assistantship – Primus GeMS Project (≈50,000 CZK)
<i>Awarded a research assistantship on the Primus GeMS project for outstanding research.</i> | 2020–2021 |
| • Excellence Scholarship at Charles University
<i>Awarded for grades in the top few %.</i> | 2021 |
| • Bronze Medal – International Mathematics Competition (IMC)
<i>Bulgaria</i> | 2020 |
| • Winner (top 10) – National Mathematical Olympiad
<i>Ranked among the national top 10 in the high-school mathematical Olympiad in Slovakia.</i> | 2016 |

INVITED TALKS

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| OCIS (Online Causal Inference Seminar)
<i>Invited speaker at the online causal inference seminar (the largest seminar in the field).</i> | Online Feb 2026 |
| Data science seminar
<i>Invited speaker at SANKEN+RIKEN - Data science and AI lab at University of Osaka.</i> | Osaka, Japan Dec 2025 |
| CHUV BDSC Seminar
<i>Invited speaker at the Clinical Data Science Group, CHUV University Hospital.</i> | Lausanne, Switzerland Sep 2025 |
| UAI 2024 Workshop on Causal Inference in Time Series
<i>Invited speaker; one of seven invited speakers including Miguel Hernán and Gustau Camps-Valls.</i> | Barcelona, Spain Jul 2024 |

CONTRIBUTED TALKS

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| Shimizu lab seminar
<i>Speaker at Shimizu lab seminar at SANKEN, University of Osaka.</i> | Osaka, Japan Dec 2025 |
| Data Driven Decisions and Inference
<i>Speaker at a weekly internal research seminar hosted by Lihua Lei, Stefan Wager, and Jann Spiess.</i> | Stanford, USA Nov 2024 |
| Bin Yu Group Seminar, Berkeley's Casual Causal Seminar and PhD Students Seminar
<i>Presentations on methodological aspects of causal discovery and extreme value theory.</i> | Berkeley, USA Oct 2024 |
| Causal Science Center Conference
<i>Contributed talk on causality and extremes.</i> | Stanford, USA Oct 2024 |
| CMStatistics 2023
<i>Speaker on causal discovery.</i> | Berlin, Germany Dec 2023 |
| EuroCIM: European Causal Inference Meeting
<i>Speaker on causal discovery.</i> | Oslo, Norway Apr 2023 |
| BIRS: Combining Causal Inference and Extreme Value Theory in the Study of Climate Extremes
<i>Speaker at a Banff International Research Station workshop</i> | Kelowna, Canada (online) Jun 2022 |
| Robust Conference
<i>Speaker: presentation on causality in extremes of time series.</i> | Prague, Czech Republic Jun 2022 |
| CMStatistics 2021
<i>Speaker: presentation on causality in extremes of time series.</i> | London, UK (online) Dec 2021 |
| Valpred Workshop
<i>Speaker: presentation on causality in extremes of time series.</i> | Aussois, France Oct 2021 |
| Extreme Value Analysis Conference
<i>Speaker: Presentation on causality in extremes of time series.</i> | Edinburgh, UK (online) Jun 2021 |

POSTERS AT CONFERENCES/WORKSHOPS

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| CLear conference at Broad Institute of MIT and Harvard
<i>Poster on counterfactual inference</i> | Cambridge, USA April 2026 |
| Causal identification and discovery workshop at Newton Institute
<i>Poster on causal discovery</i> | Cambridge, UK March 2026 |
| Causal Science Center Conference
<i>Poster on a conference hosted by Stanford Causal Science Center.</i> | Stanford, USA Oct 2024 |
| EGU24
<i>Poster on causal inference and extremes presented in the section "Many shades of causality"</i> | Vienna, Austria Apr 2024 |
| Causality in Extremes Workshop
<i>Poster at the University of Geneva workshop on causality in extremes.</i> | Geneva, Switzerland Feb 2024 |

TEACHING & SERVICE

- **Trojsten NGO - Volunteer Educator and Event Organizer** *Bratislava, Slovakia, 2016–2022*
 - Delivered 80+ math lectures to high-school students, trained national math Olympiad winners, and organized math competitions with 200+ participants.
- **Lead Teaching Assistant** *University of Lausanne*
 - **Optimization Methods** (2021–2023): Tutorials (lecture-type exercise sessions), exam preparation, mentoring.
 - **Forecasting I** (2021–2025): Course coordination, grading, and advanced project mentoring.
 - **Forecasting II** (2021–2026): Tutorials, project supervision and consultations for MSc students.
 - **Risk Analytics** (2023–2025): Tutorials, exam preparation, grading, development and supervision of applied case studies for dozens of projects.
- **Peer Review**
 - Annals of Statistics, JRSS B; JASA, Scandinavian journal of statistics, ICLR, Statistics and Probability Letters; Behaviormetrika; REVSTAT; Data Mining and Knowledge Discovery
- **First Committee Member**
Expertise Center for Climate Extremes (ECCE, UNIL)
 - First member of a PhD committee at a newly established research center; responsible for outreach and seminar & event organization.
- **Seminar Coordination**
HEC-DO Seminars
 - Primary coordinator: managed speaker outreach, budgeting and travel, publicity, and post-talk meetings.

SKILLS

- **Programming:** R, Python, $\text{\LaTeX}$; primarily use R for academic work.
- **Languages:** English (fluent), Czech (fluent), French (upper intermediate), German (basic), Slovak (native)

SELECTED CODE & REPOSITORIES

- **ITE_prediction_cross_world** 
Causal ITE prediction under cross-world assumptions
- **structural-restrictions-in-local-causal-discovery** 
Structural restrictions in local causal discovery
- **Granger-causality-in-extremes** 
Causality in extremes of time series
- **Extreme_treatment_effect** 
Extrapolating causal effects into extreme treatment domain
- **causal_cpcm** 
Conditionally parametric causal models

OTHER ACTIVITIES

- **Ultimate Frisbee – World Championships** *2016-ongoing*
National Representation
 - Represented Slovakia at the U20 World Championships
 - Also competed for the Lausanne club *Flyhigh* at the Swiss National Championships
- **Cipher Competition Series** *2020*
Main organizer of an event with 250+ participants.